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Review article

Review on impacts of low sulfur regulations on marine fuels and compliance options

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ABSTRACT

As the shipping industry grows, emissions of pollutants also increase. The international maritime organization sets regulations to reduce the environmental impact of shipping. The new IMO 2020 regulations limit the sulfur content in marine bunker fuels to 0.5 wt%. This review covers the details of regulations set for each emission type and methods used to control emissions from ships. Different conventional and non-conventional desulfurization methods are discussed for treating bunker fuel to satisfy the requirements set by the IMO. It also discusses the compliance options using non-petroleum fuels such as liquefied natural gas, alcohol-based fuels, hydrogen, ammonia, biofuels, and solar power. The advantages and disadvantages of each of these alternatives are discussed. Currently, the use of low sulfur residual blends is becoming a short-term solution to phase out high sulfur heavy fuel oils. Liquefied natural gas, methanol, hydrogen, and ammonia seem to be viable candidates to comply with the global sulfur cap. Green ammonia, as a zero-carbon bunker fuel is anticipated to meet both the GHG and sulfur reduction targets beyond 2030.

1. Introduction

The marine industry is growing rapidly in response to globalization. Nowadays, cargo ships are the main means for exporting and importing commodities worldwide [1]. Over 50,000 ships are trading internationally [2]. More than 80% of global trade is transported through international shipping [1]. Cargo ships have powerful engines that can operate at over 75 MW. Container vessels can store up to 20,000 containers and bulk carriers can transport over 300,000 tons of products. In 2019, ships transported 11.08 billion tons of goods [3]. This requires a great demand of fuel, close to 640,000 tons per day [4]. The marine residual fuel demand is approximately 49.5% of the global residual oil [5].

Considering the energy used per ton of cargo per kilometer, shipping is the most energy-efficient mode of transportation. Overall, shipping allows for an efficient, dependable, and cost-effective trade. Nonetheless, as the marine industry grows, emissions of pollutants into the air increase. When petroleum fuels are combusted in ship's engines, they emit the air pollutants such as oxides of sulfur, nitrogen, and carbon [6]. The International Maritime Organization (IMO), which is a specialized agency of the United Nations, regulates pollution from ships through the International Convention for the Prevention of Pollution from Ships (MARPOL). The new IMO regulation enforced in 2020 has caused worry in the shipping industry as it limits the sulfur content in marine fuels to 0.50 wt%. The shipping industry must consider their options to comply while maintaining the same output levels. As the regulations tighten, demands for low-sulfur petroleum fuels and also for alternative fuel types are growing. The objective of this paper is to review the marine fuel requirements, IMO 2020 regulations, ship-source pollutants and controls, and compliance options for the global sulfur cap with conventional and alternative fuels.

Abbreviations: ABS, American bureau of shipping; ADS, Adsorptive desulfurization; BN, Base number; DM, Distillate marine; ECA, Emission control areas; EEDI, Energy efficiency design index; EGCS, Exhaust gas cleaning systems; EGR, Exhaust gas recirculation; FAME, Fatty acid methyl ester; GHG, Greenhouse gases; HC, Hydrocarbon; HDS, Hydrodesulfurization; HEFA, Hydrotreated esters and fatty acids; HFO, Heavy fuel oil; HVO, Hydrotreated vegetable oil; IMO, International maritime organization; ISO, International organization of standardization; LNG, Liquefied natural gas; LSFO, Low sulfur fuel oil; MARPOL, Maritime pollution; MDO, Marine diesel oil; MDS, Microbe desulfurization; MEPC, Marine environment protection committee; MGO, Marine gas oil; ODS, Oxidative desulfurization; PM, Particulate matter; RM, Residual marine; RNG, Renewable natural gas; SCR, Selective catalytic reduction; SEEMP, Ships energy efficiency plan; SOLAS, International convention of the safety of life at sea; TPO, Tire pyrolysis oil; ULSFO, Ultra-low sulfur fuel oil; VLSFO, Very low sulfur fuel oil.

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Table 1
Viscosity and sulfur content of marine fuels [7].

Fuel type	ISO Category	Viscosity (cSt) (at 50 °C for Residual and 40 °C for distillate Fuels)		Sulfur Content (%)
		Minimum	Maximum	
Distillate Marine Fuels (DM)	DMA, DMB, DMX, DMZ	1.4	6	0.10–≤0.5
Residual Marine Fuels (RM)	RMA, RMB, RMD, RME, RMG, RMX	10	700	1.0–3.5
Ultra-Low Sulfur Fuel Oil (ULSFO-DM)	DMA, DMX	1.4	6	≤0.10
Ultra-Low Sulfur Fuel Oil (ULSFO-RM)	Mixed Fuel with RM category	8	6	≤ 0.10
Very-Low Sulfur Fuel Oil (VLSFO-DM)	DMA, DMX	1.4	6	≤0.50
Very-Low Sulfur Fuel Oil (VLSFO-RM)	Mixed Fuel with RM category	8	60	≤0.50
High Sulfur Heavy Fuel Oil (HSHFO)	RMA, RMB, RMD, RME, RMG, RMX	10	700	>0.5

2. Marine fuel composition and types

The traditional fuels used in ships are a complex mix of heavy hydrocarbons from petroleum. There exists a wide range of varieties in marine fuel depending on sources and the level of processing they receive. The liquid fuels used in ships can generally be divided into two main types: residual and distillate fuels. Residual marine (RM) fuels are made of the heavy residue that refined crude oil leaves behind as it is too heavy to boil. Distillate marine (DM) fuels are cleaner and lighter than residual marine fuels. In most cases, RM fuels are highly viscous and aromatic while DM fuels are paraffinic and have a low density [7]. Most RM fuels need to be blended with DM to reduce their viscosity. To classify marine fuels, the International Organization of Standardization (ISO) has set categories depending on the type of fuel and viscosity at 50 °C (Table 1). The most commonly used are heavy fuel oil (HFO) which is an RM-type fuel with high viscosity, marine gas oil (MGO), a DM-type fuel with low viscosity, and marine diesel oil (MDO), a blend of distillates with a small portion of HFO. MDO is a DM, categorized as DMB, as it has the highest viscosity of the DM category.

3. Marine fuel requirements

Each ship has different fuel requirements depending on its engine characteristics, storage capacity, and power requirement. There are two main power requirements that the fuel should cover:

3.1. Power for propulsion

It depends on the vessel and in some cases may combine combustion with electrical power. The range of power required varies greatly from around 0.15–100 MW. The large range of differences is due to the various ship sizes and their uses. For example, ferries, container ships, tankers, and cruises have different power needs.

3.2. Auxiliary power

This includes the power used in the ships that is not for propulsion. It can be used for instrumentation, heating, air conditioning, and emergency systems.

In the marine industry, there are two main types of engines used. In the four-stroke engine, also called the trunk piston engine, the crank and piston are connected by a single rod. These are used for small and medium ships [7]. The second type of engine is the two-stroke, also called the crosshead engine. It is used to power larger cargo ships and has a more complex system. Both engines can be used with residual or distillate fuels [8].

The following fuel characteristics determine what fuel will fulfill the necessary power requirement for functioning as stated above are:

3.3. Compatibility

Marine fuels are mostly blended products. It is important to ensure that components of blended fuel are compatible with each other. Additionally, issues can arise when fuel is switched constantly. This is done by ships to control certain emissions according to the regulations set. Nonetheless, incompatibility can cause damage to the engines, and cleaning them is difficult.

3.4. Stability

If stored for a long period, asphaltene aggregates in fuels may “break down”, causing the asphaltenes to precipitate. This makes the fuel unstable, and it cannot be reversed. Depending on the composition of the fuel there may be different outcomes. If the fuel is highly aromatic, asphaltene can remain suspended; but, if it is highly paraffinic, sludge can form that could block pipes and filters.

3.5. Catalytic fines

During the refining process, some metal particles of the catalyst, such as aluminum silicate, may leach out into fuels. These fine particles cause damage to the engine by wearing down the injection and pump [9]. Even after extraction, some metal particles may remain. The acceptable limit varies between 7 and 15 mg/kg depending on the engine specifications.

3.6. Ignition quality

It is also known as the calculated carbon aromaticity index (CCAI). It is calculated with the viscosity and density of the fuel. A higher number means less efficient combustion [9].

3.7. Density

The density of marine fuel helps to access its quality and type. The higher density value represents that fuel is heavier and has more carbon content. RM fuels tend to have a high density and high aromaticity while DM fuels are less dense.

3.8. Flash point

The minimum flash point of all fuels found in any vessel, with limited exceptions, is 60 °C or above. The limit was set by SOLAS (International Convention of the Safety of Life at Sea) and ISO 8217 to ensure safety against fires.

3.9. Pour point

It is the minimum temperature at which fuels should be stored to avoid wax formation. In some cases, with RM fuel, it is necessary to heat the fuel to maintain the temperature above the pour point. If the fuel stops flowing, it can block the filters and reduce the heat exchange capabilities [8].

3.10. Viscosity

It is inversely proportional to temperature changes. The viscosity of the fuel must be maintained in a certain range, depending on the engine,

to have efficient combustion.

3.11. Composition

Some of the components that are usually found in fuels and need to be monitored are: sulfur, water, carbon residue, and ash. The sulfur content is being regulated by the IMO to reduce SO_x pollution. Water in the fuel causes an energy loss and, in some cases, corrosion of the system [9]. Carbon deposits in the fuel can decrease efficiency [9]. Inorganic materials remain as residues after combustion in the form of ash and thus lead to scale formation. The limit of ash content is 0.2 wt% in marine fuels.

3.12. Lubricity

Fuels need to be lubricated to avoid damaging the combustion system. The less viscous fuels need more lubrication. The absence of sufficient lubrication between the pump plunger and casing can lead to pump failures. Lubrication is also required with rapid temperature changes to remove the heat [7]. If the viscosity of fuel is not in the range specific to the engine requirements, a suitable lubricant needs to be mixed into fuel to avoid damages.

4. The role of IMO in marine fuel regulations

The IMO is in charge of directing the development of international shipping towards security and the environment. Its responsibilities include regulating aspects of ship design, operation, pollutants disposal, and ship safety, sustainability, and efficiency. The waste produced by the maritime industry is significant and contaminates a lot. One of the goals of the IMO is sustainable economic growth [1]. Since its foundation in 1948, the IMO has set many regulations. The first standard regarding pollution was in 1973 with the MARPOL convention [10]. It included pollution by accidental and operational causes.

Recent regulations include controlling fuel standards and reducing CO, NOx, and SOx emissions. One of the most important regulations set in the MARPOL Annex VI is the emission control areas (ECA). These are areas generally by the coastline where the pollution regulation is stricter for the safety of the nearby population. The most recent regulation enforced in 2020 involves reducing sulfur content in fuels to 0.5% in the open sea and 0.1% in ECA [3]. Though IMO does not have the authority to enforce the regulations set, they have agreements with the Governments and national authorities of Member States [11]. These entities monitor and enforce the regulations as they have the responsibility of providing the IMO with reports. The IMO can then decide whether the country meets the requirements and set the necessary consequences.

5. Pollutant types, regulations and prevention

The ship emissions that contribute the most damage to human health and the environment are sulfur oxides, nitrogen oxides, and particulate matter (PM). These pollutants have caused over 64,000 premature deaths concentrated mostly in coastal areas [12]. Other ship emissions such as carbon dioxide and carbon oxide are greenhouse gases (GHG) that also have severe negative effects on the environment. The IMO regulations on these pollutants are described below.

5.1. Sulfur oxides

In the maritime industry, sulfur oxides (SOx) are released into the air through the combustion of fuels in marine engines [3]. The shipping industry contributes around 13% of the global anthropogenic emissions of SO_2 [13]. One of the effects of SOx is the formation of acid rain. When it precipitates, it harms vegetation and acidifies bodies of water which affects aquatic species. A study by Winnes and Fridell showed that ships contribute greater than 10% to the acidification of coastal areas [12]. SOx also causes human health issues such as respiratory diseases. A study made by the marine environment protection committee (MEPC) estimated that the number of premature deaths due to SOx emissions between 2020 and 2025 would be over 570,000 [3].

To prevent SOx emissions and improve air quality, the IMO began regulating the emissions of SOx in 2005 with the Annex VI of MARPOL. Since 2012, the maximum sulfur content in marine fuels was 3.5 wt%. The recent rule limits sulfur in marine fuels to 0.5 wt%. Getting fuels that follow the new sulfur limit is becoming a challenge for the shipping industry. Besides, to meet the new sulfur 2020 regulations, switching fuels is necessary when entering ECA unless the ship constantly runs on fuel with 0.1% sulfur.

Current options to meet the IMO sulfur regulations include treating the exhaust gas to prevent SOx from reaching the atmosphere or choosing a fuel that has a lower sulfur content. The first option involves the use of exhaust gas cleaning systems (EGCS), also known as scrubbers. There are wet and dry scrubbers. All wet scrubbers have a similar composition which includes an exhaust gas cleaning unit, a wash water treatment, and sludge handling. The exhaust gas arrives at the exhaust gas cleaning unit where it is mixed with either seawater or freshwater [14]. The mixture is then treated to remove the SOx, NOx and solid particles, and the wash water is then discharged. The formed sludge is kept until ships reach the shore and then disposed safely.

For wet scrubbers, seawater (or open) scrubbing is the most commonly used as the wash water can be returned to the sea. Additionally, the acidity of the SOx is neutralized by the alkalinity of the water. The disadvantage of this type of scrubber is its effect on water quality due to the discharge. Freshwater (or closed) scrubbers have extra chemicals added to make the wash water alkaline. Most of the water is then recycled and only a small portion is discharged. This type of scrubber is usually used in closed bodies of water because any discharge could damage the ecosystem. The efficiency of a freshwater scrubber is greater than 90%, but the disadvantage is the space taken for storage. Hybrid scrubbers exist where they use a mixture of seawater and freshwater [14].

Dry scrubbers use adsorbents to remove the SOx from the exhaust gas. The exhaust gas passes through a solid medium, generally containing calcium, and the SOx reacts to form gypsum ($CaSO_4$). The advantage of dry scrubbers is that the byproduct formed can be reused for other purposes such as fertilizers or to make cement.

The IMO MARPOL allows for the use of scrubbers as long as the SOx content released remains under 0.5% There are regulations for the discharge waters in open scrubbers where the pH must be 6.5 or higher [3]. For closed scrubbers, the discharge must be handled ashore. This method is attractive because it allows continuing the use of the current high sulfur feedstocks including heavy fuel oil. Ships with scrubbers will not have to switch fuel types which may be more costly. Nonetheless, the installation of scrubbers is awfully expensive and a long process. It can cost up to $5 million and take up to 9 months to install [4]. Some larger ships may even need multiple scrubbers installed to achieve the SOx limit. There are around 60,000 shipping vessels operating today and only 3000 have scrubbers installed [4]. The rest of the ships must use the second option to meet the SOx limit requirement. They must change from HFO to fuels that have a lower amount of sulfur. This can be achieved by mixing HFO with fuel oils with low sulfur content (LSFO) or by using alternative fuels. These options are discussed in Section 6.

5.2. Nitrogen oxides

The shipping industry contributes to ~15% of the anthropogenic NOx emissions.[13] When NOx reaches the atmosphere, it reacts to produce ozone (O_3) in the troposphere and contributes to acid rain. Additionally, it can cause eutrophication damaging bodies of water. To regulate this, the MEPC and IMO amended MARPOL Annex VI in 2008 to reduce the emission standard of NOx and SOx. The current regulations are based on power output and the engine's speed. It applies to diesel

Table 2
NOx emission limit [15].

Regulation	NO_x Limit	Revolutions per minute
Tier I	17 g/kWh	n < 130
	$45 \times n^{-0.2}$ g/kWh	130 ≤ n < 2000
	9.8 g/kWh	n ≥ 2000
Tier II	14.4 g/kWh	n < 130
	$44 \times n^{-0.23}$ g/kWh	130 ≤ n < 2000
	7.7 g/kWh	n ≥ 2000
Tier III	3.4 g/kWh	n < 130
	$9 \times n^{-0.2}$ g/kWh	130 ≤ n < 2000
	2 g/kWh	n ≥ 2000

Note: Tier I–ships made between 1 January 2000 and 1 January 2011, Tier II–ships made on or after 1 January 2011, Tier III–ships made on or after 1 January 2016.

engines over 130 KW and is summarized in Table 2.

Table 2 shows the NO_x limits for ships in ECA. Outside of ECA, they must apply Tier II limits. Unlike SOx, the treatment options for NOx do not involve changing the fuel itself. NOx is formed regardless of the fuel type used. There are two options for the ships to comply with the regulations: reduce the formation of NOx in the engine or treat the exhaust gas after NOx has been formed to prevent it from reaching the atmosphere. In the first case, it is necessary to take into consideration engine design and modifications that could be made. A study by Seddiek and Elgohary showed that injecting special additives to the liquid fuel can reduce emissions by about 20% [6]. For the second case, the most efficient treatment is selective catalytic reduction (SCR). The reducing agent, typically ammonia, reacts with the exhaust gas. SCR can be used by low– and medium-speed engines to reduce emissions by 90% [16].

5.3. Carbon emissions

Carbon monoxide and carbon dioxide are the main products of combustion. The marine industry is responsible for 3.1% of the global annual anthropogenic CO_2 emissions and 2.8% of the global annual anthropogenic GHG emissions [13]. It was also estimated that the emissions will increase from 50 to 250% by 2050 as marine trade grows [17].

In 2011 MEPC added a package to MARPOL Annex VI called "Regulations on energy efficiency for ships" for ships above 400 gross tonnage [17]. This package contained two main sections. The energy efficiency design index (EEDI) states that all new ships must comply with the minimum energy efficiency performance levels and the ship's energy efficiency plan (SEEMP). To meet these requirements new and old ships have to improve their energy efficiency by focusing on operational optimizations. In 2018 the IMO adopted an "Initial GHG reduction strategy" for ships. Its objectives are to reduce the carbon intensity of shipping by requiring new ships to emit less CO_2 emissions per transport work (g CO_2/dwt-nm). The goal is to reduce it by 40% by 2030 and 70% by 2050 from the emissions of 2008 and eventually entirely remove them to achieve the temperature goals from the Paris Agreement [17]. They also strengthened the EEDI requirements. The baseline for the EEDI is the efficiency of ships from 1999 to 2009. The EEDI for ships from 2015 to 2019 must be 10% more efficient than the baseline [18]. For ships built between 2020 and 2024, it must be 20% more efficient and 30% more efficient for ships built from 2025.

To meet these demanding requirements, ships must reach complete combustion, which means they must improve the combustion process. Another way that the IMO recommends reducing CO_2, is by operational optimizations such as planning the voyages around weather, implementing speed optimizations, and arriving in port on time. Additionally, a combined SCR system can be used to reduce CO and HC as well as NOx. The first stage oxidizes CO and HC, and the second stage reduces NOx emissions.

Table 3
Reduction methods for each emission type [6].

Component	Reduction Method	Potential Reduction
NOx	Selective Catalytic Reduction (SCR)	95%
	Emulsification	20–25%
	Humid Air	70%
	Engine tuning	50–60%
	Exhaust gas re-circulation	10–30%
SOx	Fuel Switching Process	60–90%
	Sea water scrubbing, Exhaust below water line	Up to 95%
CO_2	Energy Management	1–10%
PM	Electrostatic filters	Up to 85%

5.4. Particulate matter

Particulate matter (PM) includes any mixture of solid or liquid particles that are found in the air. They can be emitted by construction, fires, swept into the air by the wind, or formed due to reactions of chemicals found in the air. PMs include a broad array of pollutants, but the different sizes cause different effects on the environment. If they are 10 μm or under, they are easily inhaled into the lungs and cause the most harm to human health. Some of the effects include lung function deterioration, asthma, non-fatal heart attacks, and premature death of people who suffer cardiovascular or respiratory diseases [19]. PMs also have negative effects on the environment. They can be swept up by the wind and dispersed to lakes, forests, crops, and other ecosystems. PMs change the acidity and nutrient balance in bodies of water and can damage forests, and crops. Fine particles under 2.5 μm can cause visibility impairment [19].

PMs can be formed as secondary products from the emissions of SOx and NOx. According to the American Bureau of Shipping (ABS), by regulating the emission of those reactants, the IMO is also regulating the PM levels. To reduce the emissions of PM, the IMO specifically limits the amount of sulfur in the fuels used [20].

It has been found that PM emissions are greater when using HFO as there is higher sulfur content and thus more SOx emissions [12]. When higher temperatures are used, more NOx is produced, and thus more PMs are emitted. With lower loads, soot and hydrocarbon PM are higher. To reduce PM emissions, it is important to monitor fuel consumption, loading, and temperatures of combustion. By switching to LSFO, PM emissions are also decreased [6]. Another option used is catalyst filters which catch the particles before they reach the environment. Table 3 summarizes the pollutants and current methods used to reduce them by the shipping industry.

6. Compliance options

This section addresses how to meet the IMO requirements with different fuel types. Even though fuel alternatives focus on SOx reduction, in most cases there may be a reduction in the other pollutants as well. Any new alternative fuel that is developed must fulfill the fuel requirements discussed in section 3 as well as be scalable to large sizes.

6.1. Conventional fuels with low sulfur content

6.1.1. MGO and MDO

As mentioned earlier, MGO and MDO are distillate fuels that are derived from bunker fuel through processes such as vacuum distillation, thermal cracking, hydrodesulfurization and catalytic cracking. These processes also remove sulfur. The sulfur content in MGO ranges between 0.1% and 1.0% and meets the new IMO regulations. The MGO type fuels are commercially available but tend to be more expensive than other fuel options.

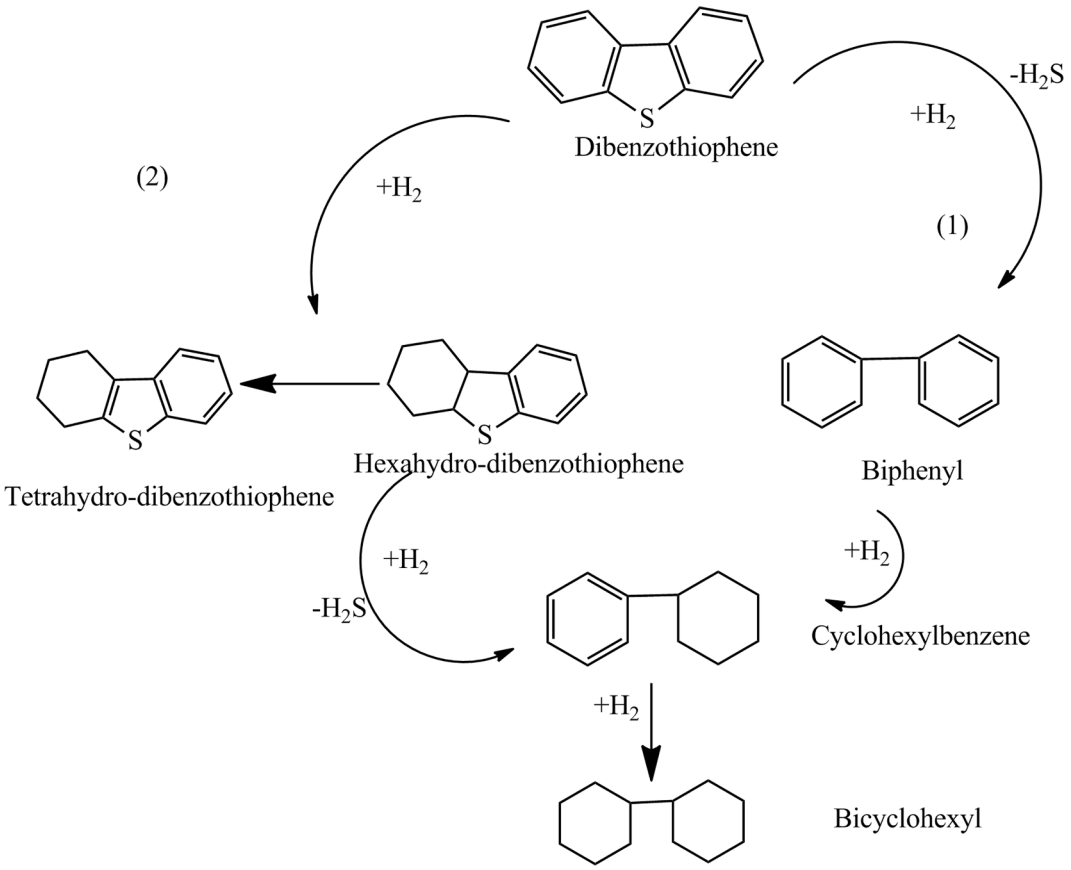


Fig. 1. Possible pathways of HDS of DBT compound, Adopted from Reference [23] with the permission from Elsevier.

6.1.2. Low sulfur fuel oils

This category includes conventional RM fuels that contain low sulfur content, also called heavy sweet fuels/crudes. Very low sulfur fuel oil (VLSFO) refers to fuels with a sulfur content of <0.5%. Heavy sweet crudes can be found in many places including Australia, Nigeria, and Brazil. These can be used with little to no treatments and are also ideal for blending for low sulfur marine fuels. VLSFO can also be obtained by treating residual oils to remove the sulfur or blending it with fuels that contain less sulfur. HFO can be blended with ULSFO, MGO, hydro-treated residuals, heavy fractions from hydro-crackers, and lighter hydro-treated fractions. Ultra-low sulfur fuel oil (ULSFO) has a sulfur content of 0.1% or less. This type of fuel is a refined product of residual oils. It is primarily used in ECA or to mix with HFO to form VLSFO.

Some issues that must be considered when using treated or blended VLSFO and ULSFO are the reduced lubricity, the increased pour point (wax formation), and incompatibility due to their high levels of paraffin. The lubricity issues can be handled by adding lubricants. As mentioned in Section 3, lubricants prevent frictional damage between the engine parts as well as control the heat levels. Nonetheless, lubricants are also important to control the pH in the engine. When fuel is burned, the sulfur in the fuel becomes SO_2 and SO_3. The SO_3 can react with the moisture in the air to form sulfuric acid (H_2SO_4). If the temperatures are maintained high, the acid does not condense. However, in slow-steam engines that use lower temperatures, cold corrosion of H_2SO_4 damages the engine over time. To neutralize the acid, chemical additives are added to lubricants. The base number (BN) of the lubricant indicates its acid-neutralizing capabilities. The higher the BN the more acid the lubricant can neutralize. With HFO, the BN of lubricant must be high, at around BN70 to BN140 in some cases [7]. But these levels of alkalinity can harm the engine if used with VLSFO. Besides, if the BM is too high, ash deposits containing calcium carbonate and other materials may form which are abrasive to the engine's interior. With more vessels switching to VLSFO, the BN of the lubricant must be closer to 40, but a new balance must be found to avoid the formation of deposits. In ECAs where ULSFO is used, the BN must be between 15 and 25. The challenge for the shipping industry is to find suitable lubricants that can be used under the new regulations.

Another aspect that must be considered is the high pour point that the VLSFO could have due to mixing with the residual oil. If the temperatures are too low, there could be wax formation which can clog the engine. It is necessary to analyze the compatibility of the blended fuel to ship engines. For example, RM fuels are very aromatic and DM fuels tend to be more paraffinic due to hydrotreating. By mixing these, some compatibility issues may arise. When blending fuels, if the viscosities and densities are too different, incompatibility issues may also arise. This indicates that mixing some HFO and ULSFO or VLSFO can cause problems. Mixing RM with DM is also not advised without testing beforehand. Similarly, fuels from different sources tend to have little compatibility. Compatibility issues can harm the filters as asphaltenes can be suspended and form sludge. When switching between fuels, the tanks need to be cleaned correctly to avoid residue incompatibility. With the new sulfur cap, it seems to be a challenge to find blended fuels that are compatible as more and more blends are expected to be produced.

Even though there are some fuels with naturally low sulfur content, their availability is limited. Considering the average demand for marine fuels is 640,000 tons per day [4], it is necessary to produce more low sulfur fuel through desulfurization. To meets the current sulfur requirement, around 250,000 tons per day of fuel oil need to be desulfurized to distillate fuels [5]. There are various desulfurization methods available to tackle this issue but not all are viable for the new demand [21]. Additionally, most methods reduce a specific sulfuric compound, but not all the sulfur compounds in the oil. New methods are being developed to improve the efficiency of desulfurization of crude oil. Some of the desulfurization methods that can be used to desulfurize bunker

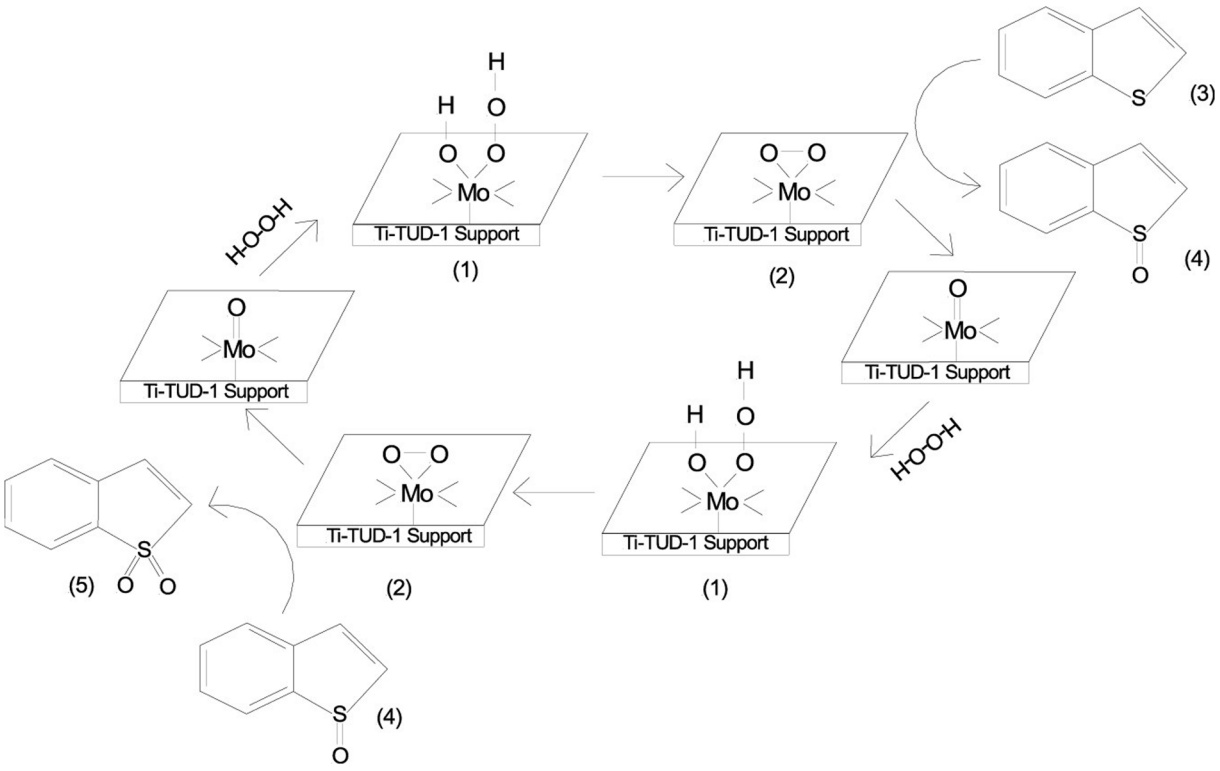


Fig. 2. Mechanism of ODS with a Keggin-Type Molybdenum HPA-Loaded Ti-TUD-1 Catalyst, adopted from reference [26] with permission from American Chemical Society.

fuels are briefly discussed below.

6.1.3. *Hydrodesulfurization*

Hydrodesulfurization (HDS) is the most common method used in the industry today. It involves reacting residual oil with hydrogen in the presence of a catalyst (usually $NiMoS/Al_2O_3$ or $CoMoS/Al_2O_3$) at high temperatures (up to 400 °C) and pressure (up to 100 atm). The mechanism of HDS with DBT is shown in Fig. 1.This method has been used to remove sulfur from HFO, and it is also used as a refining process to produce MGO and MDO [7]. HDS is not as efficient when removing more complex cyclic sulfur compounds such as dibenzothiophene (DBT) and its alkyl substitutes, particularly 4, 6-dimethyldibenzothiophene (4, 6-DMDBT). Removal of these refractory compounds from heavy fuel oil is essential to meet the new requirements. HDS of heavy fuel oil is considerably expensive because it requires high hydrogen pressures, high hydrogen flow rates, and high temperatures to remove above mentioned refractory sulfur compounds. This high-pressure process leads to huge capital investment. The disadvantages of using HDS for upgrading heavy fuel oil include increased catalyst fouling and pressure drop [22]. HDS can also saturate the aromatic hydrocarbons and thus increases the paraffin content and the pour point. Besides, the release of hydrogen sulfide gas from hydrotreaters causes unwanted pollution.

6.1.4. *Extractive desulfurization*

In this process, extractants (solvents) are used to extract sulfur compounds through liquid–liquid extraction. The solvent used is selected depending on the organosulfur compound present in the crude oil to ensure that sulfur compounds are selectively miscible with the solvent without co-extraction of other hydrocarbons. The solvent selected must have a higher boiling point than the sulfur compounds. This method is attractive because it does not require any extreme conditions and no chemical reactions that affect the fuel composition [21]. This makes it less expensive than other methods such as HDS. The disadvantages of the extraction are the complicated process of selecting a solvent and getting the crude oil to a low viscosity to aid in blending [21]. Besides, extractive desulfurization shows low efficiency with heavy oils and becomes more expensive because the solvent must have a lighter boiling point than the sulfur compounds. Additionally, other hydrocarbons with similar polarities as sulfur compounds are co-extracted by the solvent, causing a loss in fuel volume [24].

6.1.5. *Oxidative desulfurization (ODS)*

It is a two-step desulfurization process. The first step is oxidizing the sulfur compounds into corresponding more polar sulfones and sulfoxides. Many different types of sulfur compounds exist in heavy fuel oil. Sulfur compound that has a higher electron density can be oxidized more efficiently [25]. Catalysts such as zeolites, metal oxides, metal–organic framework, carbon-based compounds and polyoxometalate can be used to enhance the ODS reaction [22]. Hydrogen peroxide (H_2O_2) is a commonly used oxidant. Other options include ozone, *tert*-butyl hydroperoxide, cyclohexanone hydroperoxide, and cumene hydroperoxide. A typical mechanism of ODS with a Keggin-Type Molybdenum HPA is shown in Fig. 2.

The second step is removing the oxidized sulfur compounds from the fuel by solvent extraction. Another way of removing sulfones and sulfoxides is through thermal decomposition, as the C-S bonds are weaker in the oxidized compounds. ODS does not require extreme conditions to operate unlike HDS. Nonetheless, its commercial application is hindered due to the low efficiency in real feedstocks, carbon loss due to the removal of entire molecules along with sulfur atoms, and loss of desirable aromatic/olefinic compounds along with sulfur compounds during extraction. It has been mainly used to complement HDS to remove residual sulfur, but due to the new fuel requirements, research is being made to improve ODS through using ultrasound and ultraviolet radiation [27].

6.1.6. *Adsorptive desulfurization (ADS)*

In this method, sulfur compounds in the feedstock are adsorbed on



Fig. 3. Adsorptive desulfurization of sulfur fuels. Adopted from Reference [22] with permission from Elsevier.

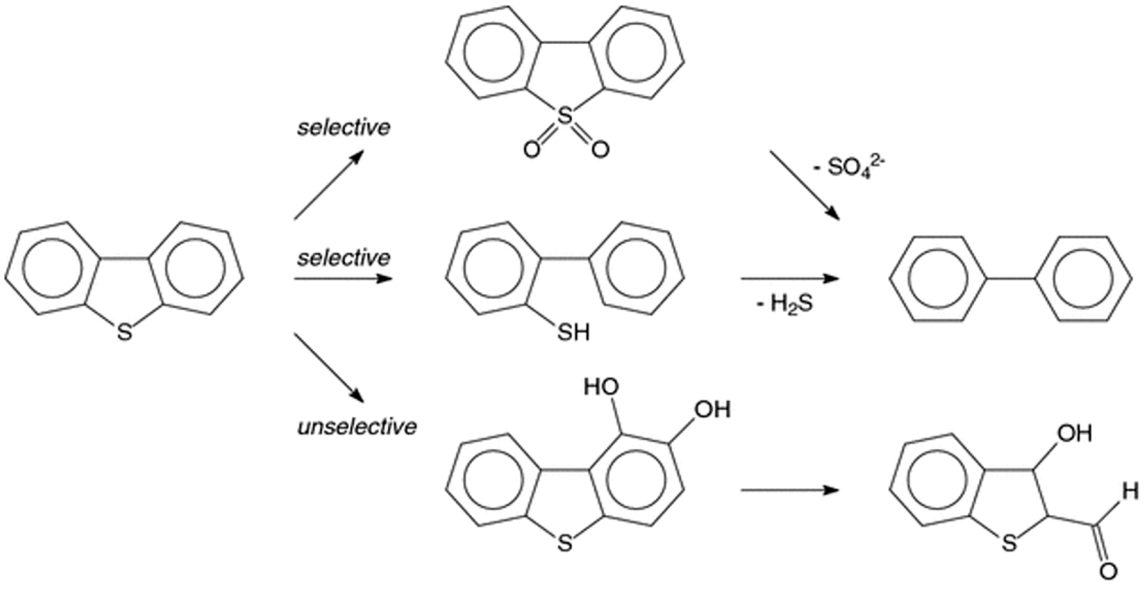


Fig. 4. Selective aerobic biodesulfurization of dibenzothiophene by Alcaligenes genus, adapted from Reference [21].

solid materials such as zeolite, activated carbon, silica-alumina, and metal–organic framework as shown in Fig. 3. This method does not require extreme conditions, which makes it relatively inexpensive and an attractive option. The efficiency of this method depends on the ability of the solid material to selectively adsorb the sulfur compounds in the presence of other compounds, and the capacity and durability [21]. The texture of the adsorbent is also important as the larger surface area, pore volume, and active sites can provide better adsorption [22]. ADS may be achieved by physisorption or chemisorption. ADS is mainly used for deep desulfurization to get ultra-clean diesel fuel and gasoline. This method has not been used for large-scale industrial applications due to the unpractical adsorption of large molecules into small pores [21]. Additionally, other compounds present in the fuel can reduce the adsorption of sulfur through competitive adsorption [28]. Nonetheless,

feasibility studies show promise for the application of ADS to real fuels [22]. More research is required to evaluate the viability of different adsorptive materials and their durability to apply ADS to larger scales.

6.1.7. Microbe desulfurization (MDS)

It is also known as bio desulfurization (BDS). The sulfur in the residual oils is metabolized by microorganisms. The temperature and pressure must be low to have the ideal conditions for this method. MDS has a relatively lower cost compared to others, due to low operational and capital costs, but has the added complication of transportation and handling of microbes. There are two types, aerobic and anaerobic MDS. Aerobic MDS using Pantoea agglomerans D23W3 shows an efficiency of 63% on heavy oils with 1.9% sulfur [29]. As shown in Fig. 4, MDS under aerobic conditions produces sulfates that can be dissolved in water and separated from fuels. For the anaerobic MDS, the most studied bacteria is desulfovibrio desulfuricans, which is more effective in removing aromatic sulfur than aliphatic sulfur [30]. Anaerobic MDS has the advantage over aerobic MDS and does not produce a sludge of oxidized hydrocarbons. Even though this method could be used directly on high sulfur fuel, it is mostly employed as a secondary process after other desulfurization processes. For example, by combining ODS with MDS there can be a combined efficiency of 91% with heavy oil [29]. The disadvantages of MDS are microbe contamination of fuels and substantial carbon oxidation due to unselective bioconversion.

It is important to emphasize that it is more difficult to remove sulfur from residual or heavy oils than that of distillates because of their high boiling point, viscosity, and sulfurs resistance. Each method described has advantages and limitations and they must be selectively applied based on fuel characteristics. Some of the desulfurization methods can be applied to remove only specific sulfur compounds. Nonetheless, due to the new regulations, these techniques must be improved to increase their efficiency to desulfurize heavier feedstocks including heavy fuel oil.

Overall, using LSFO is the most common compliance option as it requires little to no changes in the ship's engine system. The greatest issue is the availability of fuels that are below the sulfur cap. As desulfurization becomes more efficient, ship owners will get an option to use alternatives. Meanwhile, using MGO and MDO can make up for the shortfalls of LSFO production.

6.2. Alternative marine fuels

Even though LSFO is a good short-term solution, with research and new technologies other fuels that can satisfy the IMO regulations need to be developed. This involves changing the fuel itself, but it may require different engine technologies to work efficiently. Some alternative non-petroleum fuels that are being developed are discussed below.

6.2.1. Liquefied natural gas and biomethane

Liquefied natural gas (LNG) is mainly composed of methane and emits lesser CO_2 per unit of energy when burned as compared to RM and DM. Because it contains lower sulfur than fuel oils, SOx emissions are limited. Additionally, the LNG engines can be modified to reduce NOx emissions or can be fitted with an Exhaust Gas Recirculation (EGR) or SCR [31]. According to statistical analysis, LNG can reduce the NOx, SOx, CO_2 and PM emissions by 86%, 98%, 11%, and 96%, respectively when compared to HFO [32].

The United States shale gas revolution causes a significant reduction in LNG cost, making it an attractive option for the shipping industry. The US has a surplus of liquefaction capacity and an abundance of natural gas in the Permian Basin and Appalachian Basin, the top two gas fields in the US. The production costs of LNG are nearly zero and renewable penetration ensures the availability of LNG for the long term [4]. Australia and Qatar have increased their exports of LNG as well and more countries are importing the LNG fuel. Besides penetration, there is a more interesting option of obtaining LNG through renewable production. Renewable Natural Gas (RNG) is produced from refining biogas [33]. Biogas is formed from different biomass such as municipal and wastewater that are anaerobically digested by microorganisms. The methane content of untreated biogas is between 45 and 65% [33]. The biogas is processed to remove water, CO_2, N_2, O_2, and any other contaminants. The refined final product has a methane content of up to 98% [33]. Biogas provides a sustainable way of obtaining LNG.

Some smaller vessels have already implemented the change from fuel oil to LNG. In 2013, LNG was <3% of the fuel used, but by 2019, more carrier vessels had been equipped to use LNG fuel [31]. Despite sounding like the perfect solution, LNG has its shortcomings. Not many ships are equipped to burn LNG, and it requires an expensive installation. New vessels are being designed with LNG engines, but older models do not have that option. The LNG system is complex and requires a lot of space; besides the specialized engine, there must also be containment systems for storage and process systems for extraction and conditioning. For those ships that have oil fuel engines and want to make the change, retrofitting them to LNG engines is expensive. Installing a scrubber would require less space than switching to an LNG engine. Some additional risks that must be considered that are not as prominent with oil fuels are the high energy in tanks, the explosion by leakage, and the required low temperature of LNG. Also, the arrangement of the LNG systems to avoid hazards and the training of the crew are relevant aspects that must be addressed.

Another aspect that shipowners must consider, is the loading facilities at the ports. Most ports do not currently provide LNG loading facilities due to their high costs and the limited amount of LNG using ships. Nowadays more ports are considering making the change, but only a few are currently providing LNG bunkering, mostly in Northwest Europe. Vessels that use LNG as fuel are limited to trajectories near these ports. The benefits of LNG have been debated by many researchers [34]. Even though the CO_2 emissions are noticeably reduced, LNG is mainly made of methane so if there is any leakage at any point from the extracting process to the burning of LNG as a fuel, the global warming effect may be worse than the CO_2 released by burning conventional fuels. Despite the discussed disadvantages, LNG is seen by the marine industry as an attractive solution to meet the IMO regulations [35]. By developing renewable LNG– a carbon–neutral energy source, the marine industry not only can meet the IMO sulfur regulations but also helps to achieve its goal of reduction of GHG emissions by at least 50% by 2050, relative to 2008 levels.

6.2.2. Alcohol-based fuels

Nowadays, some vessels use a blend of fuel oil with an alcohol additive. With the new regulations, the proportion of alcohol is increasing, and oil is decreasing. The benefits of using alcohol blended fuels are the high-octane number and the lower sulfur emissions. Nonetheless, alcohols have a lower energy density than conventional marine fuels and require an additional volume of fuel and storage space. Alcohols have a lower flashpoint than fuel oils, which require additional safety considerations.

Bioethanol is one of the most used transportation biofuels. It is produced from renewable feedstock such as sugar cane, sugar beets, and corn through fermentation with microorganisms. Due to the high demand for these feedstocks for the food industry, ethanol can also be produced from lignocellulosic feedstocks, but this route is not yet commercialized due to its high cost. Marine fuels are only allowed to contain up to 10% of ethanol (E10) as the engines are designed to withstand only E10. When using E10 fuel, ethanol may cause swelling of non-metallic components [36]. It can also form layers of ethanol and water, which can lead to corrosion and harm the engine [37]. Additionally, there are complications when it comes to transporting and storing E10 fuel.

With the new IMO regulations, new engines are being developed to use a higher ethanol content, but reliance on methanol as a marine fuel is evolving faster because it can be produced from a wider variety of

feedstock including fossils, natural gas, coal, and renewable resources. Because of the availability and low cost of natural gas, methanol is predominantly produced through steam reforming of natural gas into syngas followed by catalytic conversion of syngas over a Cu/ZnO catalyst. Producing methanol from natural gas does not provide any valuable environmental benefits as opposed to producing from biomass. However, technologies for bio-methanol are still being developed to be used at larger scales. One of the reasons why methanol is a great option compared to others is that it can be used in the same engines used for heavy fuel oil with minor modifications [38]. Some dual-methanol engines are being developed where a small amount of diesel is mixed with methanol to aid ignition. Although it is more common to have an M85 or a fuel with lower methanol content for ships, M100 is used in some vehicles [39]. Methanol is emerging as the future of the marine industry. Nonetheless, methanol is toxic which requires additional considerations for the safety of the crew.

6.2.3. Hydrogen fuel cells

Hydrogen fuel cells are being developed as an option for vessels. A fuel cell is a device that uses electrochemical reactions to convert chemical energy from a fuel into electrical energy. They are composed porous anode, that receives the fuel, and a cathode, that receives an oxidant, separated by an electrolyte layer. A redox reaction occurs at the electrodes that produce an electric current. Different fuel cell types have been developed. Alkaline fuel cells use hydrogen and oxygen as their main components [40]. They require low temperatures but require catalysts to assist the reaction. When the hydrogen is split, the proton passes through the electrolyte layer while the electron generates electricity. Sulfuric and phosphoric acid fuel cells have good stability and low costs and use normal air and hydrogen as well as acids. They require a temperature of about 200 °C and a catalyst. Proton exchange membrane cells use a solid electrolyte made of polymer and hydrogen and air for their inputs. It was developed for transportation, but it is still being improved as the energy produced is not sufficient. Direct methanol fuel cells use pure methanol in the anode. Because of the lower electrochemical activity of methanol as compared to hydrogen, the efficiency of these fuel cells is low [40]. Molten carbonate fuel cells use carbon dioxide and oxygen as their main fuel and the electrolyte is made of a molten carbonate salt mixture [40]. These cells require high temperatures. Finally, solid oxide fuel cells have high efficiencies but use high temperatures to function. They use an ion-conducting oxide ceramic as the electrolyte. Even though they have the highest efficiencies of all fuel cells (50– 60%), they are expensive.

The advantages of using hydrogen fuel cells in the marine industry include that they usually have higher efficiencies than petroleum fuel and gas engines [41]. This is because the efficiencies are dependent on the electrochemical reactions and not on the Carnot efficiency that limits combustion. They also reduce the emissions of SO_X, NOx and PM which would comply with the IMO regulations. Fuel cells generally require low maintenance. It is easy to scale fuel cells to meet the power requirement of the ships. The disadvantages include the high cost of fuel cells and the hydrogen feedstock itself. Additionally, hydrogen is flammable, and it can cause explosions if not handled correctly.

Nowadays, fuel cells are used as a source of electricity and heat for various applications including propelling vehicles [40]. Hydrogen fuel cells are not ready to replace residual oils and MGO in the marine industry as more research and development are needed. Nonetheless, it is a good option for smaller vessels with defined routes as studies by Ballard Power Systems have shown that fuel cells can operate for over 30,000 h under these conditions [42].

6.2.4. Ammonia

With more options are being developed as marine fuels to satisfy the IMO regulations, ammonia (NH_3) has considerable advantages. Ammonia is produced mainly by the Haber-Bosch process using hydrogen obtained from fossil fuels. This process forms "brown ammonia" and is not sustainable due to the carbon emissions particularly during the production of hydrogen. "Green ammonia" can be produced from hydrogen (e–hydrogen) obtained from water electrolysis by renewable electricity [43]. This way some of the negative environmental impacts of this alternative fuel can be reduced but the cost of hydrogen production through electrolysis is higher than that of steam reforming [44,45].

Ammonia is the second-largest synthetic inorganic product produced in the world [46]. The annual worldwide ammonia production is around 180 million tons [47]. Of this amount, the majority is used in fertilizers (about 80%) and the rest is applied to a variety of industries including nutrition, healthcare, plastics, and NOx control systems in the transportation industry. The marine industry is looking at ammonia as a compliance option because it does not contain carbon that causes GHG emissions. Besides, ammonia is more energy-dense than liquid hydrogen on a volume basis [48]. It can be used at close to ambient conditions and thus easier to handle and store than liquid hydrogen [49].

Additionally, ammonia can be used as fuel both in combustion engines and in fuel cells [47,50]. Since ammonia has a low combustion rate it required a combustion promoter such as hydrogen, alcohols, diesel, and residual oils. This approach demands dual-fuel storage systems. Depending on the dual fuel system, GHG emissions might occur, reducing the benefits of using ammonia as a marine fuel. Ammonia releases high NOx emissions on combustion. But it can be overcome by using selective catalytic reduction systems [51]. More research must be done to make ammonia as a cost-effective option for marine fuel. Some initiatives have been taken to develop efficient spark-ignition (SI) and compression ignition (CI) engines for this low auto-ignition fuel but have not been commercialized yet [52,53]. In the long term, technological advancement in water electrolysis and renewable electricity is expected to reduce the cost of green ammonia and thus it will become the major zero-carbon bunker fuel for achieving shipping industry future GHG reduction targets along with sulfur [54].

6.2.5. Biofuels

Biofuels are a fuel alternative that has great potential to reduce GHG emissions. Biofuels are produced from biomass such as forest and agricultural residues, plant-based sugars, vegetable oils, and algae. Compared to petroleum fuels, biofuels contain significantly lesser amounts of sulfur and nitrogen and thus produce lower SO_x and NO_x emissions [55]. Because biomass is renewable, biofuels can become sustainable fuels for the marine industry. Biofuels can be used as standalone or blended with bunker fuels. Some biofuels that can be used in the maritime industry are briefly discussed below.

6.2.6. Biodiesel

It is also known as fatty acid methyl ester (FAME), produced from vegetable oil or animal fat by transesterification. It can be used as a drop-in fuel in marine diesel engines without modification of engines [56,57]. It is often blended with marine diesel as per ISO 8217:2017 specifications. The advantages of biodiesel include its good ignition and lubricity properties along with low emissions of SOx, CO, PM, and GHG. As a disadvantage, biodiesel can clog the filters if used under 10 °C and damage the engines. To overcome this problem, the fuel tanks can be heated or additives such as methanol ethanol, kerosene can be blended [58]. Currently, biodiesel is produced commercially using vegetable oils. Vegetable oil accounts for around 80 % of biodiesel production costs [59]. The high cost of vegetable oils makes biodiesel expensive than fossil diesel.

6.2.7. Renewable diesel

Renewable diesel (also called "green" diesel and drop-in diesel) is a hydrocarbon-type fuel produced from biomass sources such as lipids (vegetable oils, animal fats, greases, and algae) and cellulosic material (crop residues, woody biomass, and dedicated energy crops). It meets the ASTM D975 specification in the United States and EN 590 in Europe.

Table 4
Characteristics of biodiesel and renewable diesel.

Biodiesel	Renewable Diesel
Pure fuel is called B100. It is not fully compatible with petroleum diesel engines. It requires engine modification.	Chemically equivalent to petroleum diesel. Requires no changes to fuel infrastructure or vehicle's fuel system.
Highest recommended blending level is 5% ("B5") in conventional diesel.	Can be used "neat" or blended in any proportion in diesel engines.
Poor oxidation stability and cold-flow properties.	Superior cold-flow properties — suitable for very cold weather conditions.
It is less stable due to the presence of olefinic bonds.	It is more stable due to the absence of olefinic compounds.
It has lower energy content than petroleum diesel due to the presence of around 10% oxygen.	No oxygen, it has higher energy content per volume compared to biodiesel.
Produces NOx emissions, due to the presence of oxygen in the esters.	No nitrogen and sulfur organic compounds – lower NOx and SOx emissions than petroleum diesel.

Unlike biodiesel, which is an ester and has different chemical properties from petroleum diesel, renewable diesel is chemically the same as petroleum diesel, composed of hydrocarbons. These fuels are compared in Table 4

There are three promising pathways for producing renewable diesel (Fig. 5). One is hydroprocessing of lipid biomass. The second pathway involves gasification of lignocellulose to produce syngas (carbon monoxide and hydrogen) and then utilizing the Fischer-Tropsch (FT) reaction. The third route is pyrolysis or liquefaction of lignocellulose to bio-oil and then upgrading through hydroprocessing.

Currently, the vast majority of drop-in biofuels are derived via the lipid platform which uses oil crop feedstocks such as canola (or low euricic acid rapeseed), camelina, palm oil, used cooking oil and tallow. Among the different technologies, the hydroprocessing route is at a relatively high maturity level and is commercially available for the conversion of any crop oils (lipids) into renewable diesel. The fuels produced through hydroprocessing are often referred to as hydrotreated esters and fatty acids (HEFA), hydrotreated vegetable oil (HVO) and hydrogenation-derived renewable diesel (HDRD). The renewable diesel produced through this route meets the ASTM standard and has the advantages of higher cetane numbers (>70), lower aromatic and sulfur contents than its petroleum counterpart. Besides, the cold flow properties of this fuel can be varied with market demands. As summarized in Table 5, several companies have developed proprietary technologies based on hydroprocessing. Among them, technologies developed by Neste Oil, Honeywell UOP, Haldor Topsoe and Renewable Energy Group, Inc. are commercialized.

Renewable diesel can also be produced from lignocellulosic feedstocks such as those derived from agricultural and pulp and paper residues through syngas and biocrude (bio-oil) routes as shown in Fig. 5. The biocrude route involves pyrolysis or thermal or catalytic depolymerization to produce bio-oil, and then hydroprocessing (upgrading) of biocrude into renewable diesel. Several pilot plants are operational for the production of biocrudes, however, production of a drop-in grade diesel through this route is not commercialized yet since subsequent upgrading of biocrude suffers from the short lifespan of catalysts.

The syngas route involves the combination of biomass gasification, Fischer-Tropsch (FT) synthesis and hydroisomerization, wherein biomass is gasified and after cleaning, the syngas is used for FT synthesis to produce long-chain hydrocarbons that are converted into drop-in diesel through hydroisomerization/cracking. The advantages of this route are (i) the FT portion of this route is commercialized with the syngas derived from natural gas and coal, (ii) any type of organic feedstock can be used, especially forest and agricultural residues, which are in enormous supply and currently at low cost. However, the FT technology for renewable diesel is still in its early stages of scale-up from pilot plants to commercial operability since the capital cost is still quite high. The current status of renewable diesel from lipids and lignocelluloses is compared in Table 6.

Renewable diesel production through hydroprocessing of lipids is the only commercialized process as of now. Today almost all renewable diesel available in the market is produced from vegetable oil, in a standalone hydroprocessing plant. Renewable diesel can be blended with marine fuels to meet the IMO sulfur cap regulations, but the economics of producing renewable diesel from vegetable oil is not promising due to the high cost of vegetable oils [60].

6.2.8. *Tire pyrolysis oil (TPO)*

Scrap tires are discarded into landfills where they pose health and fire hazards due to their non-biodegradable nature. The uncontrolled or open burning of scrap tires is very toxic to the environment. To manage waste tire disposal, pyrolysis is used to form an alternative fuel. The oil yield varies between 40 and 50% depending on tire types and pyrolysis methods [61,62]. The pyrolytic oil (TPO) obtained from tires typically contains C_6 to C_{24} hydrocarbons with a high calorific value of around 42 MJ/kg. As such, it can be used to displace coal and coke in many industrial applications like cement kilns, thermal power stations, and industrial boilers [63]. Pyrolytic oil cannot be used directly as drop-in

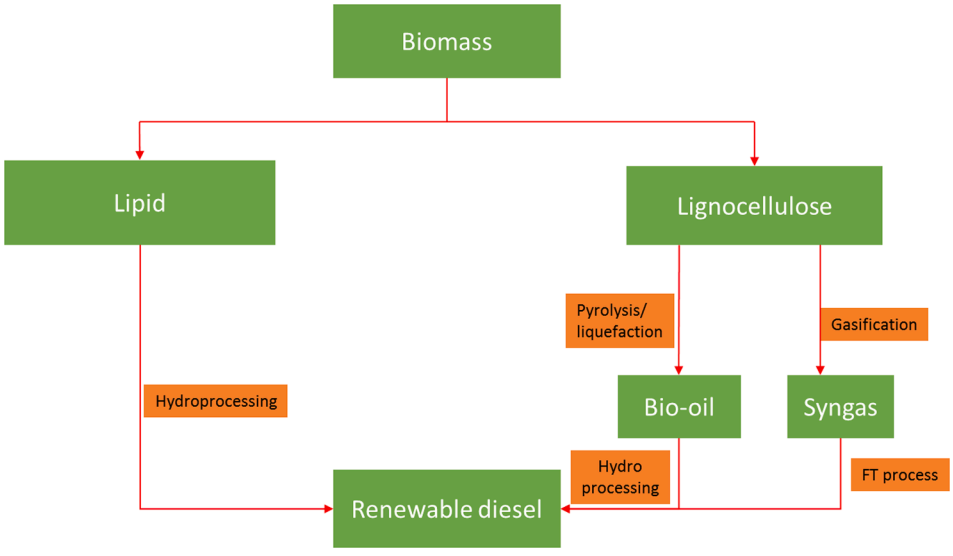


Fig. 5. Different platforms for drop-in diesel.

Table 5
Technology for renewable diesel from lipid feedstocks.

Company	Technology	Feedstock	Commercial status
Neste Oil	Hydroprocessing/ hydrocracking/ hydroisomerization	Vegetable oil and animal fat	Four commercial units: Two in Finland, one each in Rotterdam and Singapore
Renewable Energy Group, Inc.	Bio-Synfining-Hydroprocessing/ Hydroisomerization Developed by Dynamic Fuels LLC (Joint venture formed by Syntroleum and Tyson Foods)	Inedible corn oil, used cooking oil, waste vegetable oils, animal fats, greases and vegetable oils	One commercial plant in Geismar, Louisiana
Honeywell UOP	UOP/Eni Ecofining process- involves hydroprocessing, hydroisomerization	A wide range of oils and fat	Three commercial plants:(1) Diamond Green Diesel; (2) Eni S.p. A., Italy; (3)Alt Air Fuels
Haldor Topsoe	Hydroprocessing/ hydrocracking/ hydroisomerization	Vegetable oils, animal fats, waste oils	Preem Refinery in Göteborg, Sweden UPM Lappeenranta Biorefinery, Finland
Axens IFP	Hydroprocessing/ hydrocracking/ hydroisomerization	Vegetable oils, animal fats, waste oils	Total S.A planned to transform its oil refinery in La Mède, France to produce renewable diesel through Vegan technology
Forge Hydrocarbons Inc.	Supercritical water is used to remove oxygen to produce hydrocarbons, then processed to make renewable diesel. Technology developed at the University of Alberta.	Fats and oils.	Pilot-scale operation in Edmonton It is working with BIOX Corporation to co-locate a renewable diesel facility at Sombra, Ontario site.
SBI Bioenergy Inc.	Process Intensification and Continuous Flow-Through Reaction Technologies- Involves three stages: heterogeneous transesterification, deoxygenation, and rearrangement of hydrocarbon chain (Isomerization & cyclization)	FFA animal fats, yellow grease, and a wide variety of vegetable oils, tall oil, and a mixture thereof	Building of demonstration plant is in process
Applied Research Associates, Inc & Chevron-Lummus Global (CLG) & ReadiFuels	It consists of three main processes- Hydrothermal cleanup, catalytic hydrothermolysis and hydroprocessing. The first is to produce free fatty acids from vegetable oil. Catalytic hydrothermolysis is carried out at super critical water conditions to produce biocrude from free fatty acids. Biocrude is then hydroprocessed into straight chain, branched, and cyclic hydrocarbons.	Soybean oil, jatropha oil, camelina oil, carinata oil, and tung oil	First commercial plant in "engineering" for UrbanX Renewables, California.
UPM Biofuels	Uses Haldor Topsoe hydrotreating process (HydroFlex™)	Crude tall oil, which contains up to 70% of FFA.	Commercial-scale operation at UPM Lappeenranta Biorefinery, Finland.

Table 6
Status of drop-in fuels.

Parameter	Cellulosic-Based Platform	Lipid-Based Platform
Feedstock	Low-cost	Expensive
Feedstock Availability	Large supply of non-food cellulosic biomass feedstocks	Compete with the food sector for vegetable oil. Limited availability of industrial feedstocks (non-edible vegetable oils)
Process Cost	Expensive process	Expensive process
Commercial status	Only on demonstration phase, Not yet commercialized	Commercialized

transportation fuels as it contains a significant amount of polyaromatic nitrogen and sulfur compounds [61]. Upgradation of tire pyrolysis oil needs to be done to meet the IMO mandated sulfur level of 0.5 % in marine fuels. Nevertheless, TPO may be blended with distillate fuels for marine vessels propelling [64]. This is an interesting option because it removes harmful waste from the environment, Studies have shown that diesel derived from scrap tires have similar performances to standard diesel fuel [65–67]. Nonetheless, there are limited studies on using this alternative as a ship fuel.

6.2.9. Solar power

Solar power is one of the less developed compliance options. Some smaller vessels have used hybrid systems of solar power and batteries to power their trajectories [68]. Nonetheless, the photovoltaic cell panels require large areas to generate enough electricity for ships, and large storage spaces. Because of the limited onboard space availability, not all ships can adopt this method. Research is still being done to solve this problem. Some ships have developed vertical solar panels that also act like sails to help the propulsion of the vessel. One system used by NYK and Nippon Oil Corporation in their Auriga Leader project was able to provide 10% of the energy required for a car carrier vessel while in the dock. This was achieved by using 328 solar panels. A ship can produce 1.4 times of energy when it is at sea as opposed to being in the ports [68]. This is a promising option for small vessels with short trajectories, especially when used to charge rechargeable batteries. Overall, this method can be beneficial when it is used to support other fuel options [68,69]. For example, solar electricity can be used to power hydrogen fuel cells. Recently, Eco Marine Power Co. Ltd (EMP) unveiled a sailboat (EnergySail l®) that utilizes both wind and solar power.

Every fuel type has advantages and disadvantages that must be considered by each shipowner. They must select the right fit for their needs. While all the fuel options discussed are promising, most require research and development to make them efficient and economical to the marine industry. The alternate fuels can be used on smaller vessels to reduce emissions until they are ready to power massive vessels. Meanwhile, most ships will probably continue to use conventional fuels to comply with the IMO regulations.

7. Conclusions

The marine industry is rapidly growing, and societies depend on it for the transportation of goods. As the reliance on shipping grows, the pollutant emissions also increase. It is in the best interest of everyone to

make it as sustainable as possible and regulate the industry. The IMO has set strict limits on the emissions of SOx, NOx, GHG, and PM. The most recent regulation was implemented in January 2020 and it required the sulfur content of marine fuels to be reduced to 0.5% in open sea and 0.1% in ECAs. As the marine sector looks for compliance options, new fuels are being developed. The requirements are plenty for liquid fuels, including high energy density, specific viscosities, flash points, pour points, lubricity, stability, and availability. Some fuel options that have been developed are conventional oils with low sulfur content such as VLSFO or ULSFO. This is an attractive option because these do not require engine changes as they use the same systems. On the other hand, changing to biofuel types, not only reduces SOx emissions but can also reduce GHG emissions. The shipping industry can be prepared for future IMO regulations if alternative non-petroleum fuels are used. These included LNG, hydrogen fuel cells, alcohol-based fuels, ammonia, biofuels, and solar power. The disadvantage of these fuels is the need for modification to the ship's system as they have different requirements and storage necessities. In some cases, they have lower energy densities or higher production costs. These alternatives are still being researched and developed to be viable options, but for now, smaller vessels can choose these to power short voyages. Biofuels can serve as bunker fuels in the short term, however, increases in demand from other sectors may result in constraints on the supply. More research must be done to make biofuels as a cost-effective option for the shipping industry, which includes using non-food feedstocks. The changes that are coming to the marine industry will be gradual. As shipowners try to find the most efficient compliance option, they may choose to blend fuels or use hybrid systems. As of now, to maintain the level of output, most ships are relying on LSFO blends and scrubbers. In the longer term, green ammonia is anticipated to be a major zero-carbon bunker fuel as it holds many favorable features including scalability, economics, and technical and safety implications than green methanol and green hydrogen.

CRediT authorship contribution statement

Sundaramurthy Vedachalam: Conceptualization, Writing – original draft, Writing - review & editing, Project administration. **Nathalie Baquerizo:** Writing – original draft, Writing - review & editing. **Ajay K. Dalai:** Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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